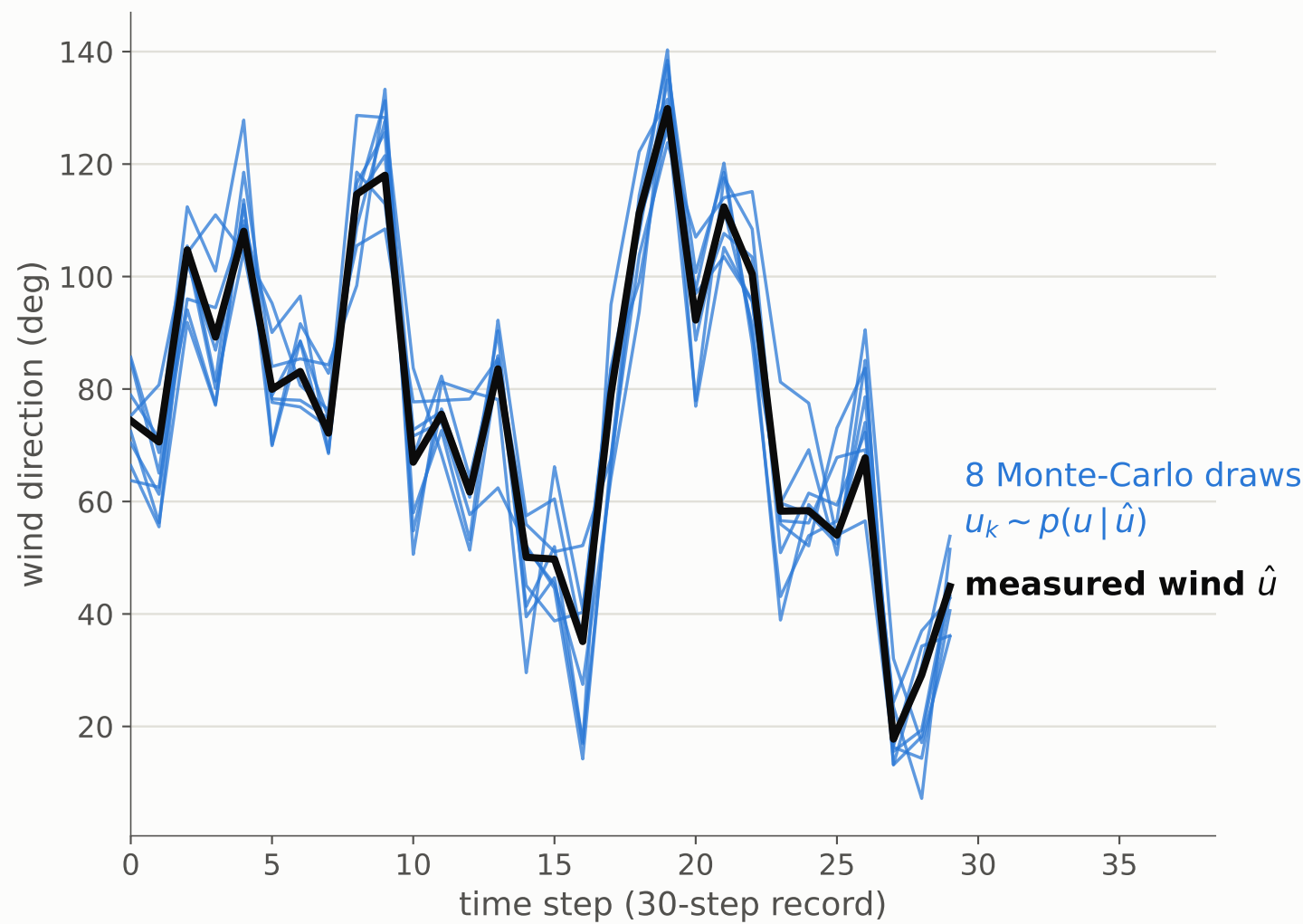


Monte Carlo: eight winds stand in for the one we couldn't measure

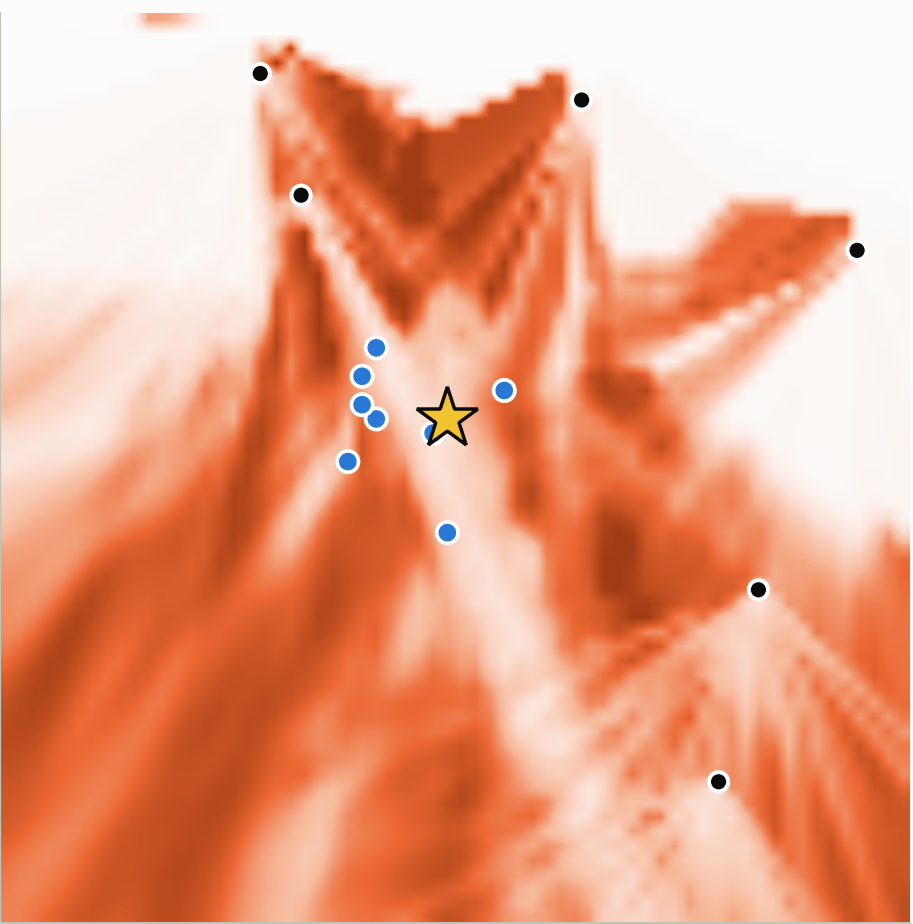
the wind-marginalization machinery on the held-out example scenario of figs 01/01.5 (6 masts, 126 kg h⁻¹); identical RNG streams to the frozen dataset, recomputed marginal checked against the stored oracle

A. One measured wind, eight simulated truths



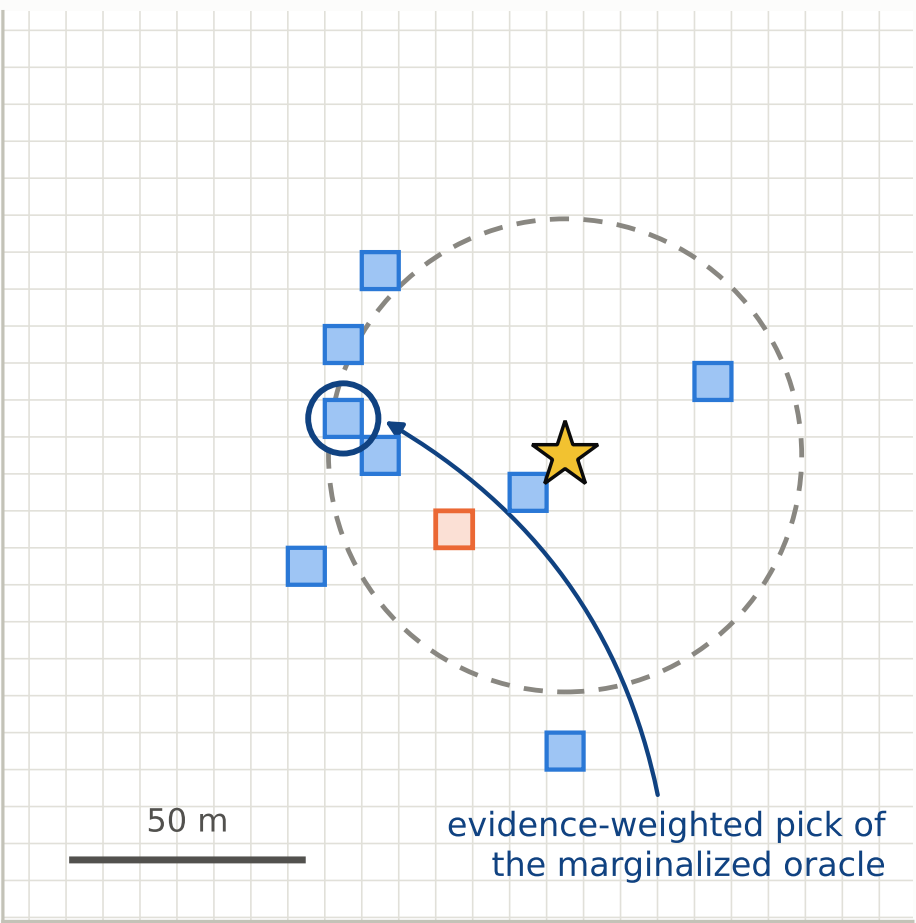
each draw re-perturbs every step by the anemometer's stated error (10° direction, 10% speed) — a sample from what the true wind could be

B. Each draw pins the leak somewhere else



dots: the single most likely cell under each wind draw — every one a near-certain posterior. Backdrop: the wind-spread input channel, the same Monte-Carlo disagreement handed to the network as a map

C. The scatter of the answers is the real uncertainty



zoom near the source: eight 7.8 m cells, 11–62 m from the truth. One wind gives false certainty; eight draws reveal the wind-limited resolution the regions must cover

What “Monte Carlo” means

Estimate an integral you cannot solve by averaging over random samples of the unknown. Here the unknown is the true wind behind the measured record:

$$p(y|c) = \mathbb{E}_{u \sim p(u|\hat{u})} p(y|c, u) \approx \frac{1}{K} \sum_{k=1}^K p(y|c, u_k)$$

The pipeline runs this estimator twice, at the same $K = 8$ budget so baseline and method are comparable:

- input maps — mean (consensus) and spread (disagreement) of the evidence over the network's 8 draws;
- physics-only oracle — average of the 8 per-draw likelihoods, weighting each draw's answer by its evidence (panel C, ring).

Payoff (2,000 scenarios × 3 seeds): the Monte-Carlo map channels shrink the median 90% region from 99 m to 74 m.